WSQ COURSE SLIDES

AI for Lean Manufacturing



WSQ Course Code: TGS-2023020425

Lean Manufacturing | PRE-OPR-4064-1.1

Version v7.0 | 19 August 2026

Tertiary Infotech Academy Pte Ltd

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TOPIC 0

Welcome and Course Administration

Two days of lean thinking, manufacturing analytics and responsible AI decision design

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Digital Attendance (Mandatory)

1 AM attendance

Use the SSG digital-attendance QR code shown by the trainer.

2 PM attendance

Complete the second attendance event after lunch.

3 Assessment attendance

Complete the separate assessment attendance event.

4 Funding gate

At least 75% attendance and competent assessment outcome are required.

TRAQOM is the digital-attendance system, not the course survey.

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About the Trainer



General Trainer

Complete before class

Name: _____

Title / designation: _____

Qualifications: _____

Areas of expertise: _____

Training and industry experience: _____

Contact: _____

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About the Trainer



Dr Alfred Ang

Principal Trainer | Tertiary Infotech Academy

- Role | Principal Trainer
- Focus | AI, process improvement and digital transformation
- Delivers | WSQ professional and technology programmes
- Approach | Evidence-led, practical and learner-centred
- Organisation | Tertiary Infotech Academy Pte Ltd
- Contact | enquiry@tertiaryinfotech.com

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LEARNER INTRODUCTION

Let's know each other

1 Role

Your role, organisation and manufacturing context.

2 Process

One process you want to improve.

3 Evidence

What data or observations are available today?

4 Outcome

What would make these two days valuable?

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LEARNING ENVIRONMENT

Ground rules

1 Participate

Share evidence, test ideas and ask questions.

2 Respect

One conversation at a time; challenge ideas, not people.

3 Safety

Do not bypass site, equipment or information controls.

4 Time

Return from breaks promptly and keep activities moving.

5 Integrity

Use AI transparently; never present generated content as verified fact.

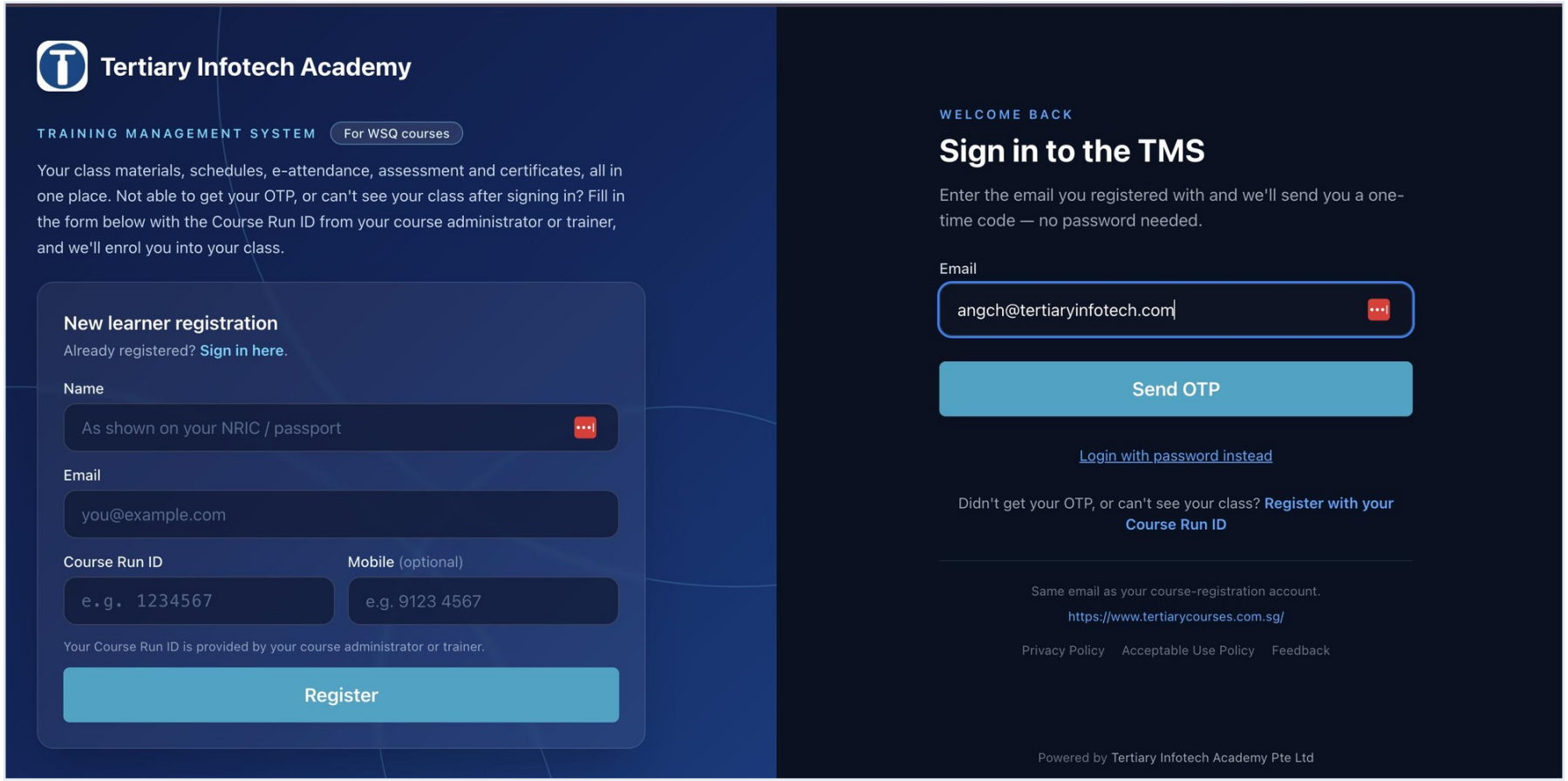
6 Confidentiality

Use the supplied scenarios and remove sensitive workplace data.

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COURSE PORTAL

Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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TWO-DAY LESSON PLAN

Course map

1 Topic 1

Lean Foundations, Waste and Flow

2 Topic 2

Inventory and Production Control

3 Topic 3

AI-enabled Lean Implementation and Maintenance

4 Activities

Ten progressive case-based activities

5 Assessment

WA (60 min) + PP (90 min), open book

6 Evidence

Worksheets, CSV analysis, checklists and pilot charter

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REGISTERED OUTCOMES

Learning outcomes

1 LO1

Identify eight types of production wastes within the organisation, benefits of lean manufacturing and steps to achieve lean performance

2 LO2

Apply lean principles to compute production capacity and TAKT time.

3 LO3

Perform ABC analysis for inventory classification using selective inventory control.

4 LO4

Calculate economic batch quantity and number of kanbans required.

5 LO5

Identify objectives of production and inventory control and types of inventories by function and condition.

6 LO6

Implement lean manufacturing tools for improved inventory flow and material flow.

7 LO7

Implement lean manufacturing in an organisation at various levels and within various areas for improvement.

8 LO8

Determine the necessity for preventive maintenance within the organisation.

9 LO9

Identify performance measurement areas in lean manufacturing and steps required for successful lean

The AI layer strengthens how learners apply the registered Lean Manufacturing outcomes.

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ASSESSMENT ADMINISTRATION

Briefing for Assessment

1 Individual work

No discussion; answer in your own words.

2 Open book

Use course slides, Learner Guide and approved materials only.

3 Devices

Phones and other unauthorised materials under the table.

4 Submission

Complete the provided document and submit on the LMS.

5 Outcome

Competent (C) or Not Yet Competent (NYC).

6 Rights

Re-assessment and appeal processes are available.

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FINAL ASSESSMENT

Assessment and funding

1 Written Assessment

8 open-ended knowledge questions | K1-K8 | 60 minutes.

2 Practical Performance

6 applied tasks | A1-A6 | 90 minutes.

3 Open book

Course materials and approved references only.

4 Funding

Attendance and competent assessment requirements apply.

The current papers preserve the original instrument, question count, codes and timing.

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Assessment flow



Complete every step in sequence; raise any accessibility need before the assessment begins.

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LEARNER MATERIALS

Course activities are evidence packs

1 Repository

A single learner-facing source for the current activity materials.

2 Self-contained folders

Each activity carries its scenario, mock data, worksheet and checklist.

3 Paired formats

Markdown supports digital work; matching PDFs support print and annotation.

4 Evidence packet

Completed worksheet, analysis output and checklist show the decision trail.

https://github.com/tertiarycourses/TGS-2023020425-AI-For-Lean-Manufacturing

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BROWSER-BASED PRACTICE

Interactive Lean and Six Sigma tools

1 **5 Whys**

https://alfredang.github.io/5whys/

2 **Fishbone**

https://alfredang.github.io/fishbone/

3 **Pareto chart**

https://alfredang.github.io/paretochart/

4 **System loop**

https://alfredang.github.io/systemloop/

5 **SIPOC**

https://alfredang.github.io/sipoc/

Use the tools to structure thinking; the evidence and decision remain yours.

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LMS OPERATING MODEL

The LMS is the controlled delivery point

1 Course record

Connects the learner to the correct registered course and cohort.

2 Current artifacts

Publishes the approved slides, Learner Guide and activity link set.

3 Assessment brief

Provides the assigned WA and Practical Performance candidate papers.

4 Submission evidence

Records the learner's completed assessment files and submission status.

https://lms-tms.tertiaryinfotech.com/

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TOPIC 1

Lean Foundations, Waste and Flow

Customer value, eight wastes, five principles, capacity, takt and problem framing

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Lean is a management system for customer value

1 Purpose

Create the value the customer needs, when it is needed.

2 Method

Expose and remove non-value-added time, effort, inventory and defects.

3 System

Link flow, pull, quality at source, people capability and continuous improvement.

4 Discipline

Use evidence and standard work; do not confuse local speed with system performance.

AI supports sensing and decisions; lean supplies the value logic and improvement method.

Source: https://www.twi-global.com/technical-knowledge/faqs/faq-what-is-lean-manufacturing

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Lean is not simply cost cutting

Lean thinking

1. Protect customer value
2. Remove causes of waste
3. Build capability
4. Improve the whole system

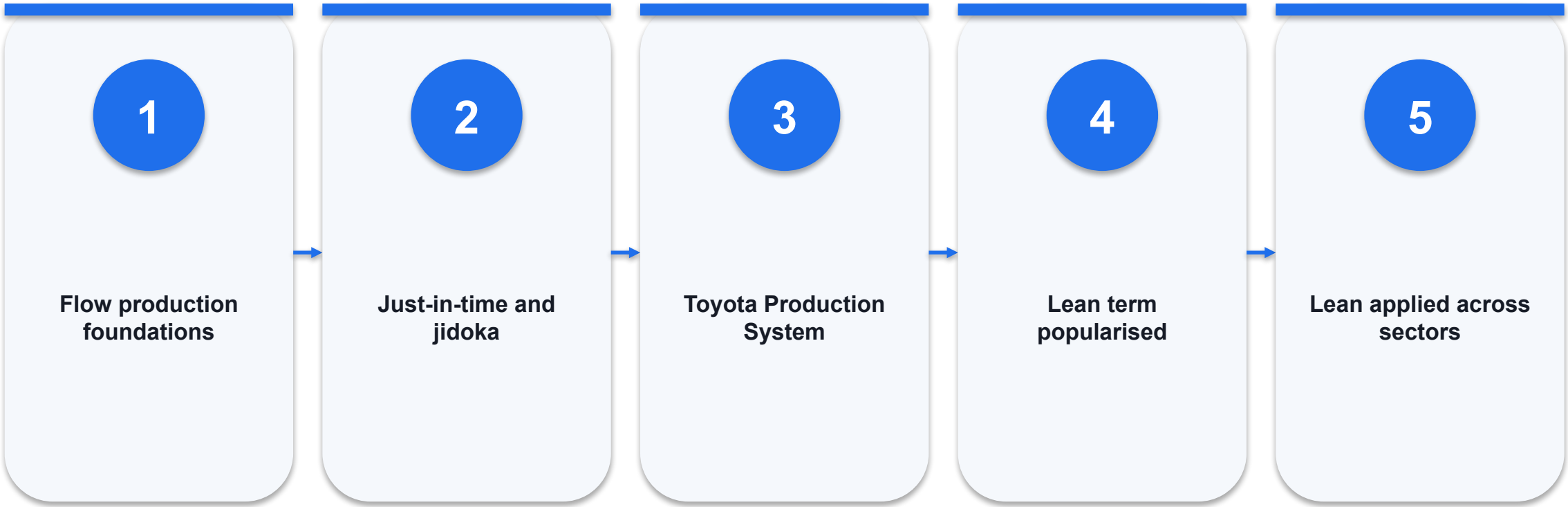
Cost cutting alone

1. Targets expense without flow
2. Can shift waste downstream
3. Often reduces resilience
4. May suppress improvement ideas

Use total-system measures before declaring a saving.

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From Toyota Production System to modern lean



The history matters because lean is a socio-technical operating system, not a toolbox.

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Customer value sets the improvement direction

1 Specific

Value is defined for a product or service, customer and need.

2 Observable

Translate needs into quality, delivery, cost, safety and service requirements.

3 Testable

A change is valuable only if it improves an agreed customer or business outcome.

4 Balanced

Safety, regulation and ethics remain constraints even when customers do not see them.

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Three activity types in a value stream

Value-adding

1. Transforms form, fit or function
2. Customer is willing to pay
3. Done right the first time

Non-value-adding

1. Pure waste: remove
2. Necessary waste: reduce or redesign
3. Always test safety and compliance constraints

Classify the activity, not the job title or department.

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The eight wastes: DOWNTIME

1 Defects

Errors, scrap, rework and inspection loops.

2 Overproduction

Making earlier or more than downstream demand.

3 Waiting

Idle people, equipment, information or material.

4 Non-utilised talent

Ideas, knowledge and capability left unused.

5 Transportation

Unnecessary movement of materials or information.

6 Inventory

Excess raw material, WIP or finished goods.

7 Motion

Unnecessary movement by people or equipment.

8 Extra-processing

Work beyond what the customer or process needs.

Source: https://www.twi-global.com/technical-knowledge/faqs/faq-what-is-lean-manufacturing

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Muda, mura and muri reinforce one another

What operators see

1. Muda: visible waste
2. Mura: uneven demand and workload
3. Muri: overburden and strain

System response

1. Stabilise demand and work
2. Level flow where practical
3. Design safe capacity margins
4. Then remove residual waste

Removing muda while ignoring unevenness and overburden can make the system brittle.

Source: https://www.twi-global.com/technical-knowledge/faqs/faq-what-is-lean-manufacturing

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Waste walk: a symptom is not yet a cause

SCENARIO

A machining cell misses shipment cut-off twice a week. WIP piles up before inspection while operators search for gauges.

SIGNALS TO INTERPRET

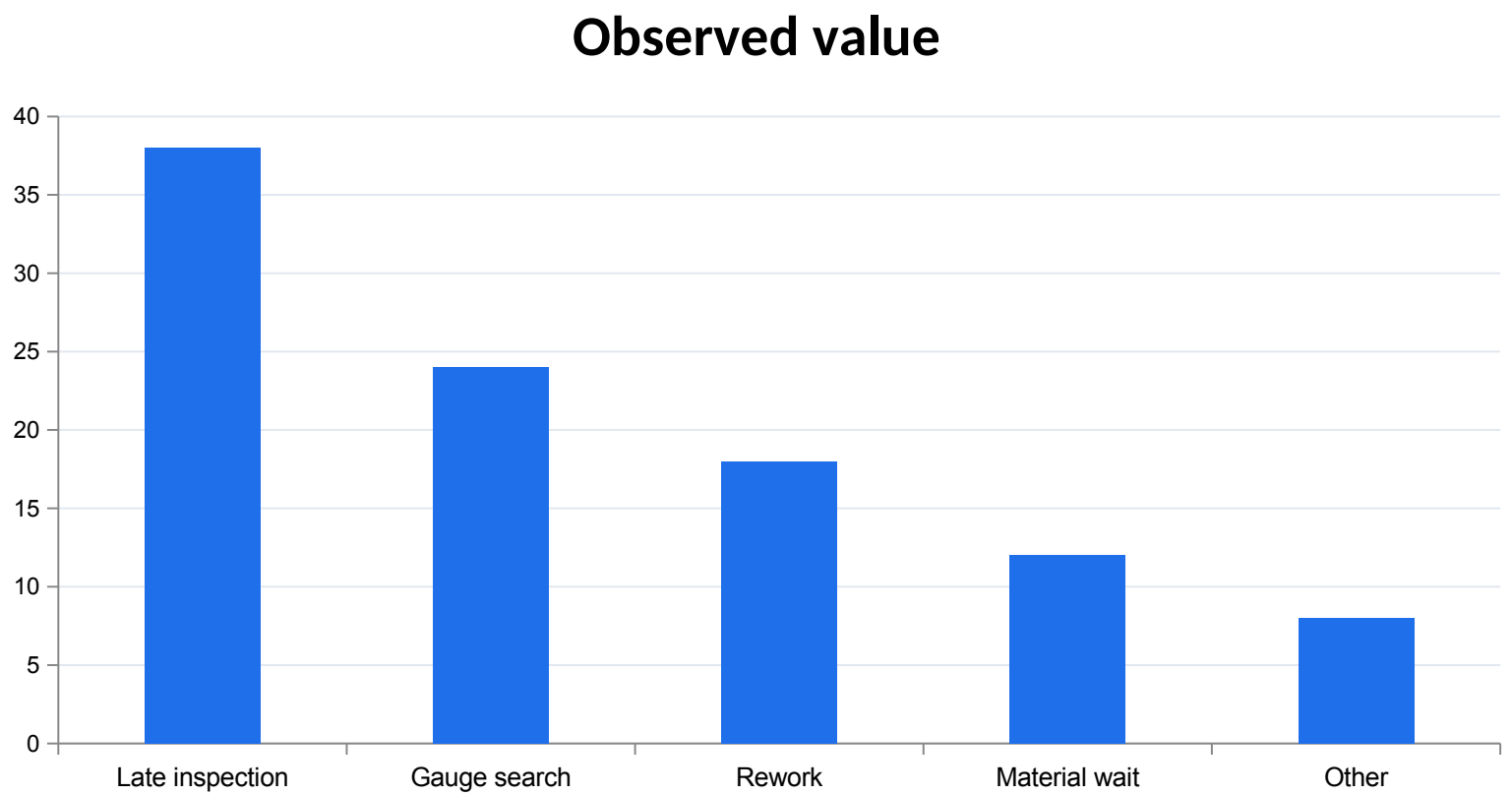
1. Waiting before inspection
2. Motion searching for gauges
3. Inventory between stages
4. Potential defects and rework

DECISION

Record observable facts first; use root-cause tools before prescribing automation.

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A Pareto view focuses the first investigation



WHAT THE DATA SHOWS

Two delay categories account for 62% of observed incidents; validate them at the gemba before acting.

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Five lean principles

1 1 Define value

Specify value from the customer's point of view.

2 2 Map the value stream

See every material and information step.

3 3 Create flow

Reduce stoppages, queues and hand-off friction.

4 4 Establish pull

Replenish from downstream demand rather than forecast alone.

5 5 Seek perfection

Continuously close the gap through learning cycles.

Source: https://www.twi-global.com/technical-knowledge/faqs/faq-what-is-lean-manufacturing

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Value-stream thinking crosses departments



Optimise end-to-end lead time and value, not the utilisation of one function.

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Current-state mapping needs facts

1 Time

Cycle, queue, changeover and lead time.

2 Flow

Material, information, rework and escalation paths.

3 Quality

First-pass yield, defect and rework rates.

4 Demand

Customer rate, mix, volatility and due dates.

5 Resources

People, machines, uptime and constraints.

6 Rules

Batch sizes, schedules, approvals and policies.

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Flow and pull solve different problems

Flow

1. Work progresses without interruption
2. Small transfer batches
3. Balanced steps
4. Visible constraints

Pull

1. Downstream demand authorises work
2. WIP is bounded
3. Replenishment signals are explicit
4. Overproduction is controlled

Use flow where processes can connect; use pull where buffers are necessary.

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Production capacity is a constraint statement

1 Rated capacity

Theoretical maximum under stated conditions.

2 Effective capacity

Expected output after planned losses.

3 Actual output

What the system achieved.

4 Bottleneck capacity

The lowest sustainable stage rate that governs the flow.

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Production rate

Production rate = good units produced / productive time

WORKED EXAMPLE

960 good units / 8 hours = 120 units per hour

MANAGEMENT INTERPRETATION

State whether downtime, breaks and rejects are included; otherwise rates are not comparable.

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Takt time aligns work to customer demand

Takt time = net available production time / customer demand

WORKED EXAMPLE

420 available minutes / 210 units = 2.0 minutes per unit

MANAGEMENT INTERPRETATION

Takt is a demand rhythm, not the observed cycle time. A process must reliably cycle at or below takt.

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Takt time, cycle time and lead time

Different clocks

1. Takt: required demand pace
2. Cycle: time to complete one unit at a process
3. Lead: elapsed time from request to delivery

Management use

1. Takt sizes capacity
2. Cycle exposes constraints
3. Lead time reveals queues and delays

Do not report one metric as a substitute for another.

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Capacity decision with variation

SCENARIO

A line has a 2.0-minute takt. Average cycle time is 1.8 minutes, but the 90th percentile is 2.4 minutes and changeovers create a morning queue.

SIGNALS TO INTERPRET

1. Average suggests capacity is sufficient
2. High-end variation misses takt
3. Changeover loss is time-dependent
4. Queueing appears despite acceptable average

DECISION

Stabilise the process and changeovers before buying capacity; model the full distribution, not the mean alone.

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Where AI complements lean foundations

1 Sense

Detect anomalies, defects and emerging constraints from data.

2 Predict

Estimate failure, demand or quality risk earlier.

3 Recommend

Rank options while exposing assumptions and confidence.

4 Generate

Draft work instructions, summaries and improvement documentation.

5 Learn

Feed verified outcomes back into standards and models.

AI should shorten the learn-and-improve loop without hiding process ownership.

Source: https://www.sciencedirect.com/topics/engineering/lean-manufacturing

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Automation, analytics, machine learning and GenAI

Deterministic

1. Automation follows explicit rules
2. Analytics describes and diagnoses
3. Outputs are repeatable for the same inputs

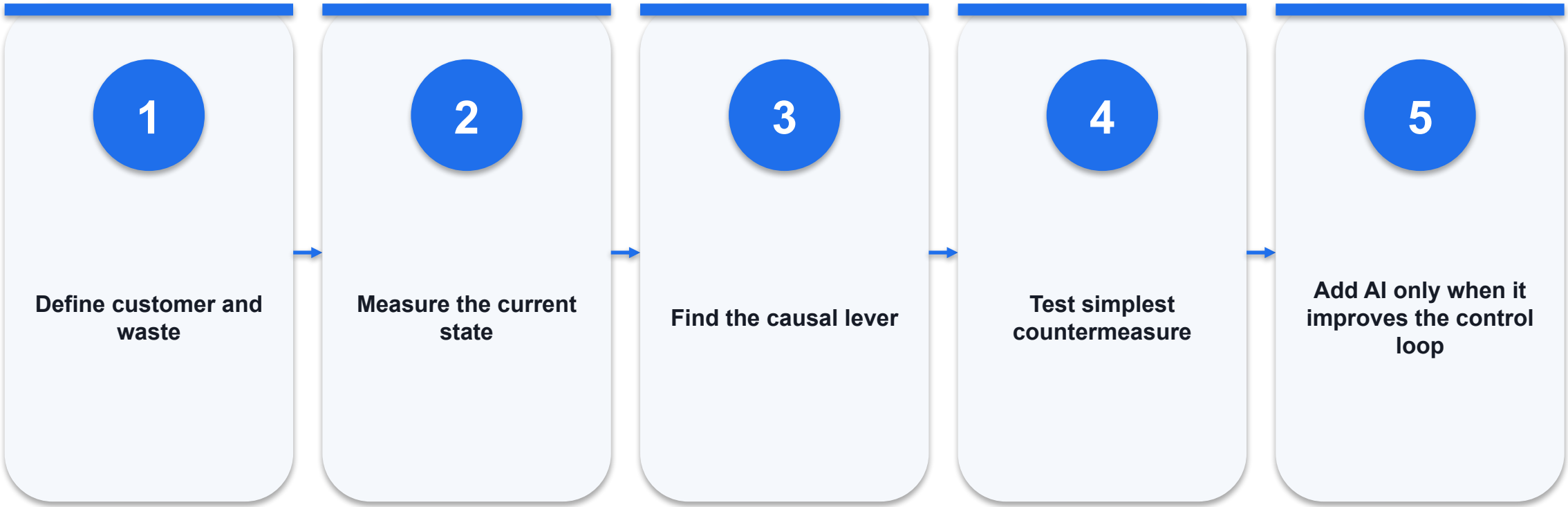
Probabilistic

1. ML predicts from patterns
2. Computer vision classifies images
3. GenAI produces new text, images or code
4. Outputs need confidence and review

Choose the least complex method that reliably solves the problem.

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Lean problem framing before an AI pilot



Start with a process problem and measurable outcome - never with a model looking for a use case.

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AI opportunity: inspection bottleneck

SCENARIO

Manual inspection takes 45 seconds per part and creates a queue. Defects are visually observable but rare.

SIGNALS TO INTERPRET

1. Potential: computer-vision triage
2. Constraint: class imbalance and image quality
3. Risk: false negatives release defects
4. Need: human review and traceable evidence

DECISION

Pilot AI as an assistive screen with conservative thresholds; retain human disposition authority.

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FOUNDATION | K1 K2 A1

Activity 1: Waste Walk and AI Opportunity Scan

THE CHALLENGE

Orion Precision Components has rising lead time in a CNC-inspection-packaging value stream. Learners use 30 observation records to distinguish facts, waste symptoms and candidate AI support.

PURPOSE

Identify the eight wastes, separate necessary from pure waste, and frame one evidence-based improvement opportunity without jumping to technology.

LEARNER OUTPUT

Completed waste register, Pareto summary, problem statement and AI/non-AI countermeasure screen.

Tools: Observation CSV; Pareto tool; 5 Whys tool; worksheet | Detailed procedure: Learner Guide and activity folder

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Activity 1 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 1: evidence and acceptance

1 Question

Which three wastes contribute most to delay?

2 Question

What evidence is still missing before causal analysis?

3 Question

Which countermeasure should be tried before AI?

4 Question

What decision could AI support, and who owns it?

5 Acceptance

All observations are classified with a rationale.

6 Acceptance

Pareto totals reconcile to the CSV.

7 Acceptance

Problem statement contains no proposed solution.

8 Acceptance

AI option names a decision, data, human owner and risk.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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Activity 2: SIPOC and Value-stream Decision Map

THE CHALLENGE

A medical-device subassembly moves through order entry, kitting, assembly, inspection and release. Handoffs and information delays are not visible in the current flowchart.

PURPOSE

Define process scope with SIPOC, classify value-added and non-value-added work, and produce a current-state decision map.

LEARNER OUTPUT

SIPOC, current-state value-stream worksheet, flow/pull improvement hypothesis and evidence plan.

Tools: SIPOC tool; system-loop tool; worksheet | Detailed procedure: Learner Guide and activity folder

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Activity 2 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 2: evidence and acceptance

1 Question

What is the customer-defined output?

2 Question

Which steps transform the product correctly the first time?

3 Question

Where does information wait?

4 Question

What feedback loop sustains the queue?

5 Acceptance

SIPOC boundary is one end-to-end process.

6 Acceptance

Customer requirements are measurable.

7 Acceptance

Value classification follows the three lean tests.

8 Acceptance

Future-state hypothesis includes a measure and constraint.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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INTERMEDIATE | K4 A4

Activity 3: Capacity, Takt and Bottleneck Analysis

THE CHALLENGE

Orion must ship 210 good units in a 420-minute net shift. Four process steps have different cycle-time distributions, downtime and yields.

PURPOSE

Calculate takt, effective capacity and constraint risk, then select an improvement experiment based on facts rather than averages alone.

LEARNER OUTPUT

Capacity table, takt comparison, bottleneck decision and countermeasure experiment.

Tools: CSV; calculator; worksheet | Detailed procedure: Learner Guide and activity folder

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Activity 3 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 3: evidence and acceptance

1 Question

What is the takt time?

2 Question

Which process constrains good-output capacity?

3 Question

How does variation change the conclusion?

4 Question

What should be stabilised before capacity is purchased?

5 Acceptance

Takt is 2.00 minutes per good unit.

6 Acceptance

Effective capacity calculations show units and assumptions.

7 Acceptance

Constraint decision considers variation and yield.

8 Acceptance

Countermeasure changes one causal factor with a defined test.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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Topic 1 recap

1 Knowledge

Eight types of production waste

2 Ability

K1 K2 K4 A1 A4

3 Evidence

Use the completed activity outputs to explain your decision.

4 Transfer

Name one workplace process where the method applies.

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Lunch Break

60 minutes | Return ready for inventory and flow

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TOPIC 2

Inventory and Production Control

Inventory purpose and types, ABC, EBQ, kanban, JIT, 5S, VSM and QC tools

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Production and inventory control objectives

1 Service

Meet demand with the right item, quantity and timing.

2 Flow

Keep material moving without starving or flooding processes.

3 Cash

Limit capital tied up in stock and obsolescence.

4 Quality

Protect identity, condition and traceability.

5 Resilience

Use explicit buffers for credible variability and risk.

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Inventory by function

1 Cycle stock

Supports normal replenishment between orders or batches.

2 Safety stock

Protects service against demand or supply variability.

3 Pipeline stock

Material moving or waiting between locations.

4 Anticipation stock

Built ahead for predictable peaks or shutdowns.

5 Decoupling stock

Buffers adjacent processes with different rhythms.

6 MRO

Maintenance, repair and operating supplies.

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Inventory by condition

1 Raw material

Not yet transformed.

2 Work in progress

Partly processed or waiting between operations.

3 Finished goods

Completed and awaiting shipment.

4 Serviceable

Fit for planned use.

5 Quarantine

Held pending inspection or disposition.

6 Obsolete or excess

No current demand or beyond useful life.

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Inventory can be protection and waste

Useful buffer

1. Absorbs known variability
2. Protects a constraint
3. Supports service commitments
4. Has an explicit owner and rule

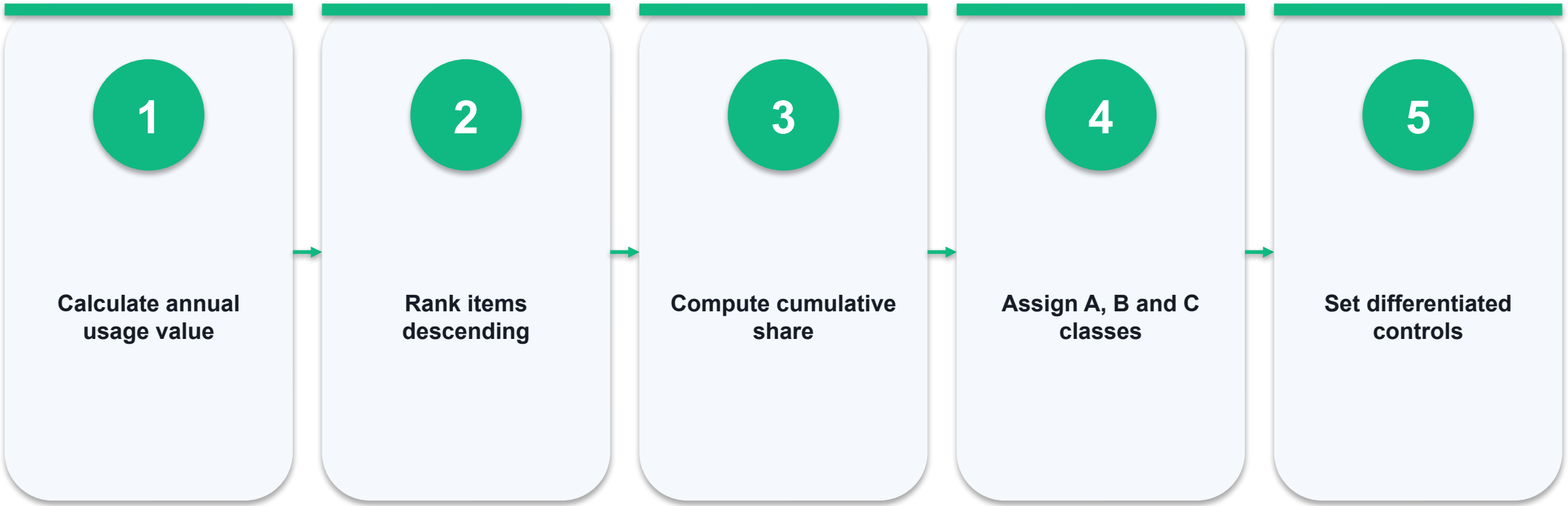
Hidden problem

1. Masks quality and downtime
2. Lengthens lead time
3. Consumes space and cash
4. Can become obsolete

Every buffer needs a purpose, size rule and exit condition.

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ABC selective inventory control



ABC prioritises management attention; it does not replace criticality, lead-time or risk analysis.

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Annual usage value

Annual usage value = annual demand x unit cost

WORKED EXAMPLE

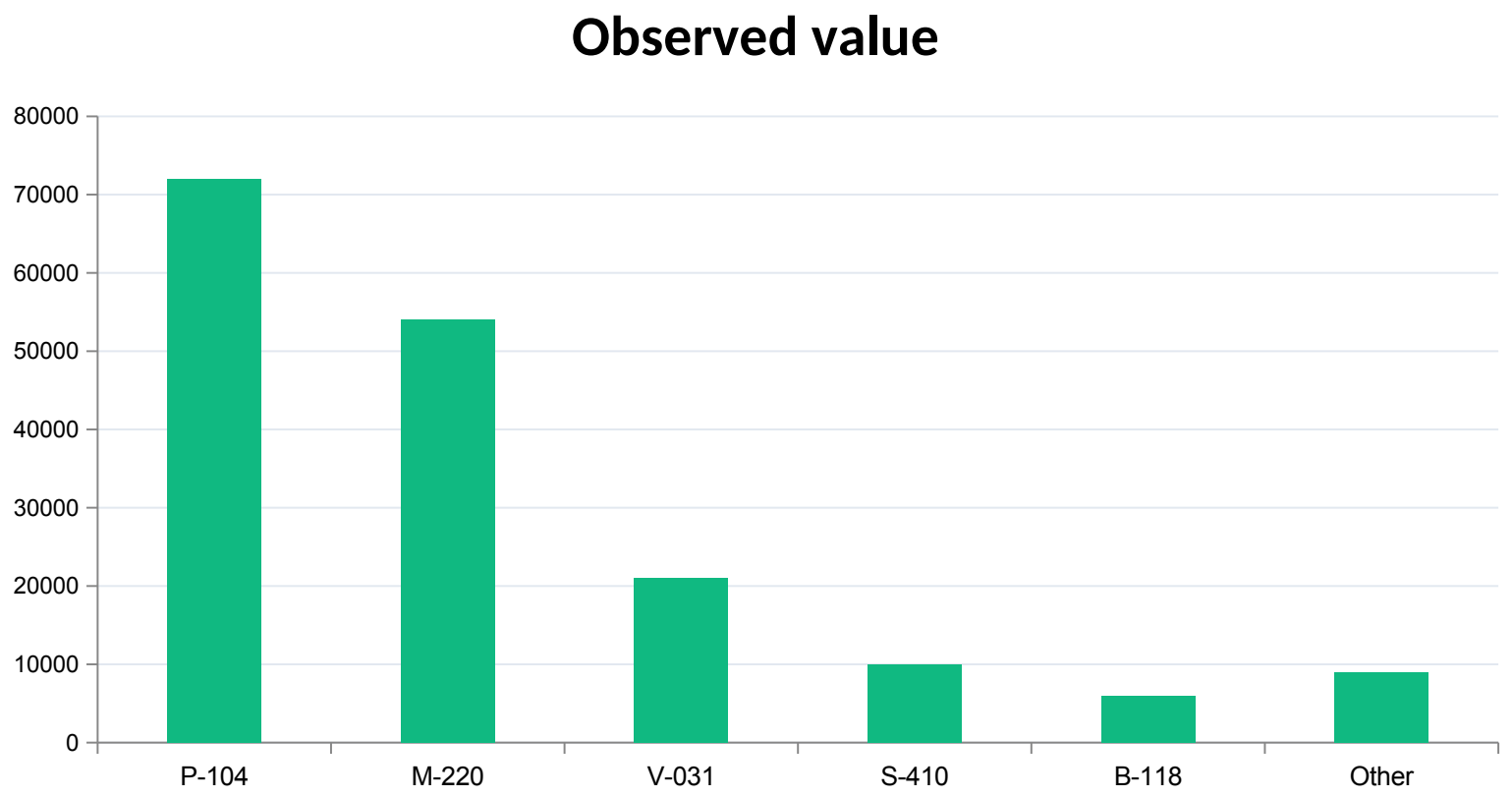
Item P-104: 4,000 units x $18 = $72,000

MANAGEMENT INTERPRETATION

Sort values before cumulative percentages; keep item identifiers as text.

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A small item set can dominate annual value



WHAT THE DATA SHOWS

The top two items represent most annual usage value and deserve tighter review and records.

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Class policy must follow the classification

A items

1. Frequent review
2. Accurate records
3. Tight approval and supplier management
4. Small controlled buffers

B and C items

1. B: periodic review and balanced control
2. C: simple controls and efficient ordering
3. Still override for safety-critical items

Use a criticality flag alongside ABC to prevent low-cost safety parts being neglected.

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Economic batch quantity balances two costs

1 Setup cost

More batches increase preparation, changeover and administration.

2 Holding cost

Larger batches increase space, capital, damage and obsolescence.

3 Demand

EBQ assumes a known production requirement.

4 Production rate

The finite replenishment rate distinguishes EBQ from basic EOQ.

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Economic batch quantity

$$EBQ = sqrt[(2DS/H) \times P/(P-D)]$$

WORKED EXAMPLE

D=12,000/year, S=$80/setup, H=$3/unit-year, P=30,000/year -> EBQ approximately 1,033 units

MANAGEMENT INTERPRETATION

Check P > D and document assumptions; revise when demand, setup or holding costs change.

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EBQ is a decision aid, not a command

Assumptions

1. Stable demand and production rate
2. Known setup and holding costs
3. No material stockout during production
4. Consistent quality

Reality checks

1. Shelf life and space
2. Supplier or machine constraints
3. Changeover improvement opportunity
4. Demand uncertainty

Lean often reduces the setup cost itself, allowing smaller economical batches.

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Kanban is a signal-and-response system

1 Signal

Card, empty bin, electronic message or controlled slot.

2 Authorisation

Permits production or movement - no signal, no work.

3 Limit

The number of cards caps WIP.

4 Feedback

Blockages and abnormal consumption become visible.

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Kanban quantity

Kanbans = demand rate x replenishment lead time x (1 + safety factor) / container quantity

WORKED EXAMPLE

500 units/day x 2 days x 1.10 / 50 units = 22 kanbans

MANAGEMENT INTERPRETATION

Round up to a whole container and review the safety factor after stabilising lead time.

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Kanban sensitivity

SCENARIO

The team wants more kanban cards because stockouts occurred during supplier delays.

SIGNALS TO INTERPRET

1. More cards hide the lead-time problem
2. A permanent buffer increases WIP
3. Supplier delay distribution is not understood
4. Container size may be poorly matched

DECISION

Contain short-term risk, then reduce and stabilise replenishment lead time before resetting the card count.

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Lean tools for material flow

1 5S

Makes location, quantity and abnormality visible.

2 Value-stream mapping

Connects material and information flow.

3 Standard work

Defines the current best-known safe method.

4 JIT

Produces and replenishes near actual demand.

5 Heijunka

Levels volume and mix where demand permits.

6 SMED

Reduces changeover so smaller batches are practical.

7 TPM

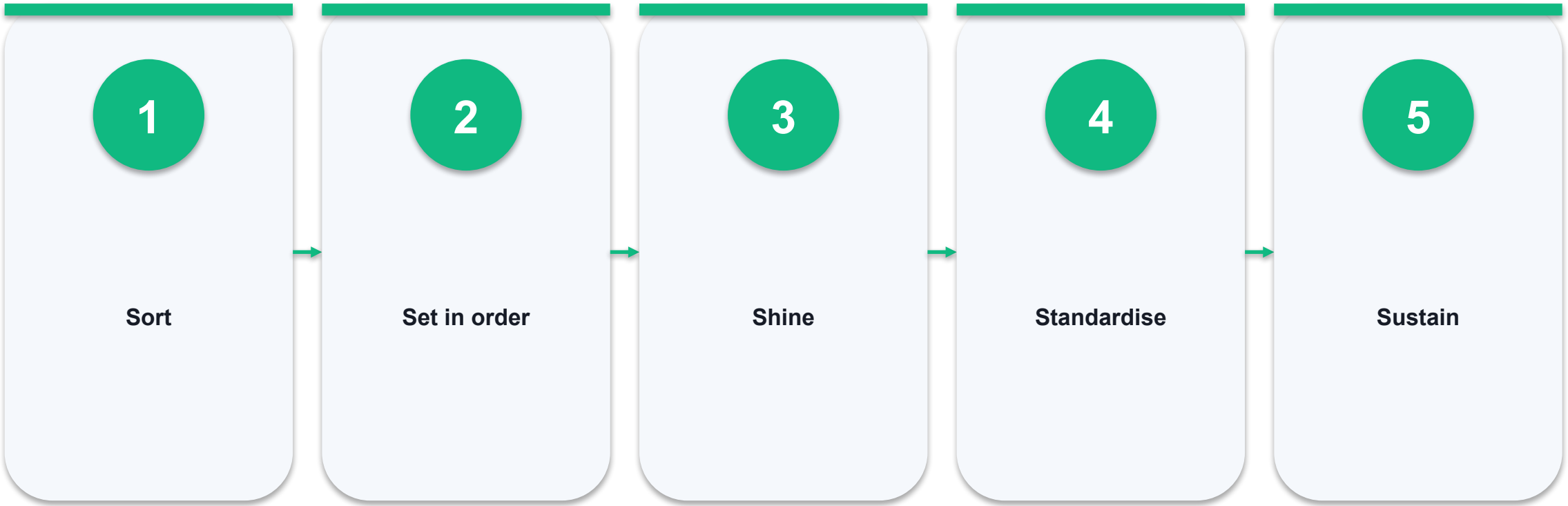
Protects equipment availability and stable flow.

8 Poka-yoke

Prevents or immediately detects errors.

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5S creates a visual baseline



5S is not a one-time clean-up; it is a control system that makes normal and abnormal conditions obvious.

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Just-in-time needs stability

Enablers

1. Reliable processes and suppliers
2. Short replenishment lead time
3. Quality at source
4. Visible demand signals
5. Flexible people and equipment

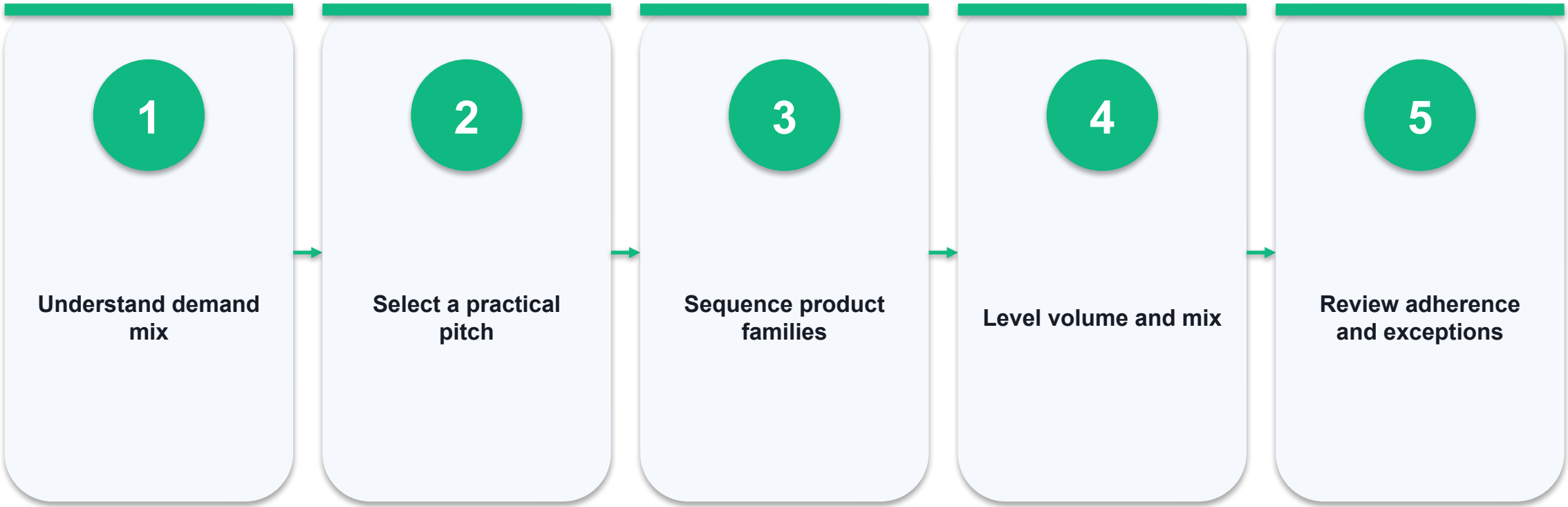
Failure modes

1. Too little buffer for real variability
2. Expediting replaces improvement
3. Local pull disconnected from customer demand
4. Risks transferred to suppliers

Design resilience explicitly; low inventory is an outcome, not the sole objective.

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Heijunka reduces unevenness



Level only what the market and process can support; do not flatten meaningful customer priorities.

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AI demand forecast versus pull control

SCENARIO

A forecasting model predicts next week's mix, while the line uses kanban based on actual withdrawal.

SIGNALS TO INTERPRET

1. Forecast helps capacity and material planning
2. Kanban authorises near-term replenishment
3. Forecast error must not inflate every buffer
4. Human planners need exception thresholds

DECISION

Use prediction for planning horizons and pull for execution; reconcile them through explicit exception rules.

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AI can improve inventory decisions

1 Demand sensing

Use recent orders, seasonality and context to update forecasts.

2 Anomaly detection

Flag unusual consumption, scrap or stock movement.

3 Supplier risk

Surface lead-time and quality patterns.

4 Replenishment recommendation

Prioritise reviews while preserving approval authority.

5 Data assistant

Summarise exceptions and draft action lists from verified ERP data.

Source: https://www.sciencedirect.com/topics/engineering/lean-manufacturing

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AI recommendation versus replenishment authority

AI may

1. Score risk
2. Forecast demand
3. Rank shortages
4. Suggest order quantities
5. Explain drivers

Human owner must

1. Validate data and constraints
2. Resolve safety and service trade-offs
3. Approve material commitments
4. Record overrides and outcomes

Never allow a language model to invent inventory balances or supplier commitments.

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ABC plus criticality plus AI

SCENARIO

A low-cost seal is class C by annual value but stops the only coating line when unavailable.

SIGNALS TO INTERPRET

1. ABC alone under-prioritises it
2. Criticality is high
3. Lead time is volatile
4. AI forecast confidence is low because failures are rare

DECISION

Use a multi-criteria policy: annual value + operational criticality + lead-time risk + data confidence.

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INTERMEDIATE | K5 K6 A2

Activity 4: ABC Inventory Classification with Criticality

THE CHALLENGE

Orion carries twelve materials with different demand, cost, lead time and operational criticality. Management currently reviews every item monthly.

PURPOSE

Perform ABC analysis, challenge value-only classification with criticality and design differentiated control policies.

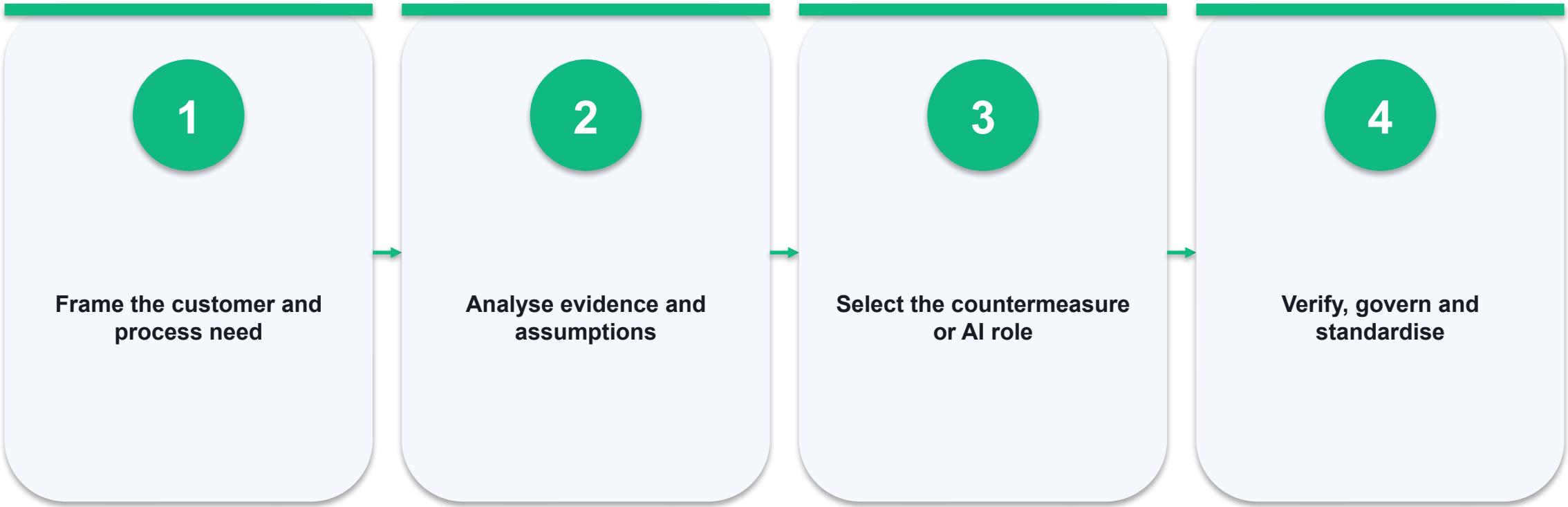
LEARNER OUTPUT

Ranked ABC table, cumulative-value chart, criticality overrides and class-control policy.

Tools: Inventory CSV; Pareto tool; worksheet | Detailed procedure: Learner Guide and activity folder

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Activity 4 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 4: evidence and acceptance

1 Question

Which items form approximately 80% of annual usage value?

2 Question

Which low-value items need an override?

3 Question

What review frequency fits each class?

4 Question

How could AI assist without auto-ordering?

5 Acceptance

Annual-value total and cumulative percentages reconcile.

6 Acceptance

Classes are based on ranked value and documented boundaries.

7 Acceptance

Criticality overrides are explicit, not hidden.

8 Acceptance

Policy differs meaningfully across A, B and C.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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INTERMEDIATE | K7 A3

Activity 5: EBQ and Kanban Control Design

THE CHALLENGE

A component family is made internally at a finite production rate. Large batches reduce setups but increase WIP; replenishment delay also creates stockouts at assembly.

PURPOSE

Calculate EBQ and kanban quantity, test assumptions and design a signal-and-response rule.

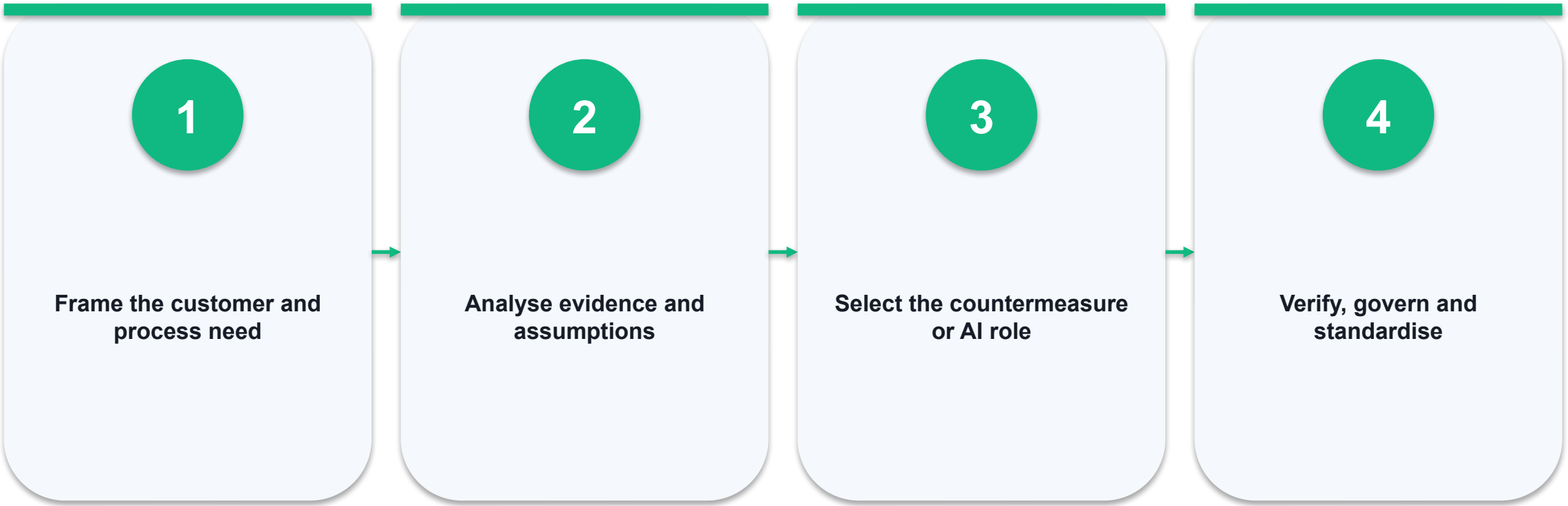
LEARNER OUTPUT

EBQ calculation, kanban quantity, sensitivity table, card/bin rule and review trigger.

Tools: Calculation worksheet; calculator; checklist | Detailed procedure: Learner Guide and activity folder

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Activity 5 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 5: evidence and acceptance

1 Question

What is the EBQ under the base assumptions?

2 Question

How does setup reduction change it?

3 Question

How many kanbans are required?

4 Question

Which assumption most needs evidence?

5 Acceptance

Base EBQ is approximately 1,033 units.

6 Acceptance

Setup reduction produces a smaller economic batch.

7 Acceptance

Kanban result is rounded up and unit-consistent.

8 Acceptance

Signal rule states authorisation, limit and exception response.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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INTERMEDIATE | K7 A6

Activity 6: Root Cause with 5 Whys and Fishbone

THE CHALLENGE

CNC-2 generates intermittent surface marks after changeovers. The first response was to retrain operators, but the problem returned.

PURPOSE

Use structured root-cause tools to separate evidence from assumptions and select a testable countermeasure.

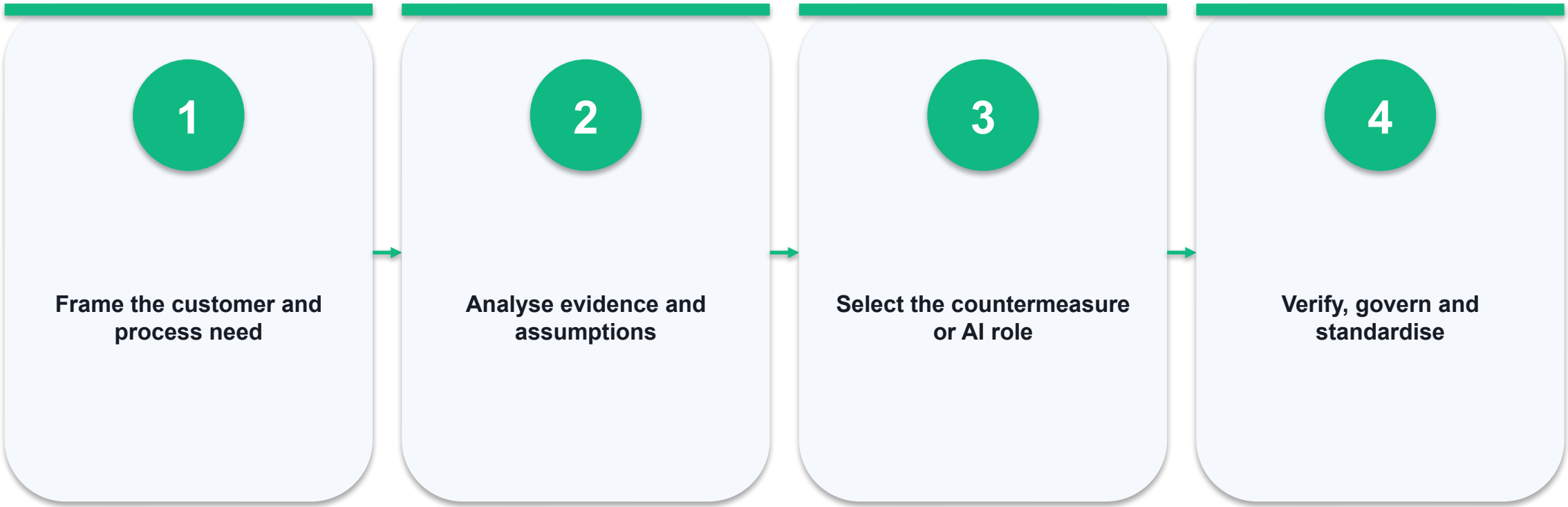
LEARNER OUTPUT

Problem statement, 5 Whys chain, fishbone map, evidence plan and countermeasure test.

Tools: 5 Whys tool; Fishbone tool; evidence checklist | Detailed procedure: Learner Guide and activity folder

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Activity 6 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 6: evidence and acceptance

1 Question Where does the causal chain rely on an assumption?	**2 Question** Which cause category is under-investigated?	**3 Question** What evidence would disconfirm the leading hypothesis?	**4 Question** How could GenAI help without fabricating a cause?
5 Acceptance Problem statement includes magnitude, location and time pattern.	**6 Acceptance** At least two causal branches are considered.	**7 Acceptance** Assumptions are visibly tagged.	**8 Acceptance** Countermeasure follows evidence and has a verification date.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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Topic 2 recap

1 Knowledge

Objectives of production and inventory control

2 Ability

K5 K6 K7 A2 A3

3 Evidence

Use the completed activity outputs to explain your decision.

4 Transfer

Name one workplace process where the method applies.

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End of Day 1

Bring your completed worksheets for Day 2

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03

TOPIC 3

AI-enabled Lean Implementation and Maintenance

Metrics, TPM, predictive maintenance, quality AI, GenAI, governance and scalable implementation

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Preventive maintenance protects lean flow

1 Reliability

Avoid breakdown-driven waiting, defects and schedule churn.

2 Safety

Control equipment deterioration before hazardous failure.

3 Quality

Keep process conditions inside proven limits.

4 Cost

Plan labour and parts instead of paying emergency premiums.

5 Learning

Use work history to remove recurring failure causes.

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Reactive, preventive, condition-based and predictive

Time and event based

1. Reactive: after failure
2. Preventive: calendar, usage or event schedule
3. Simple to govern
4. May over-maintain or miss early deterioration

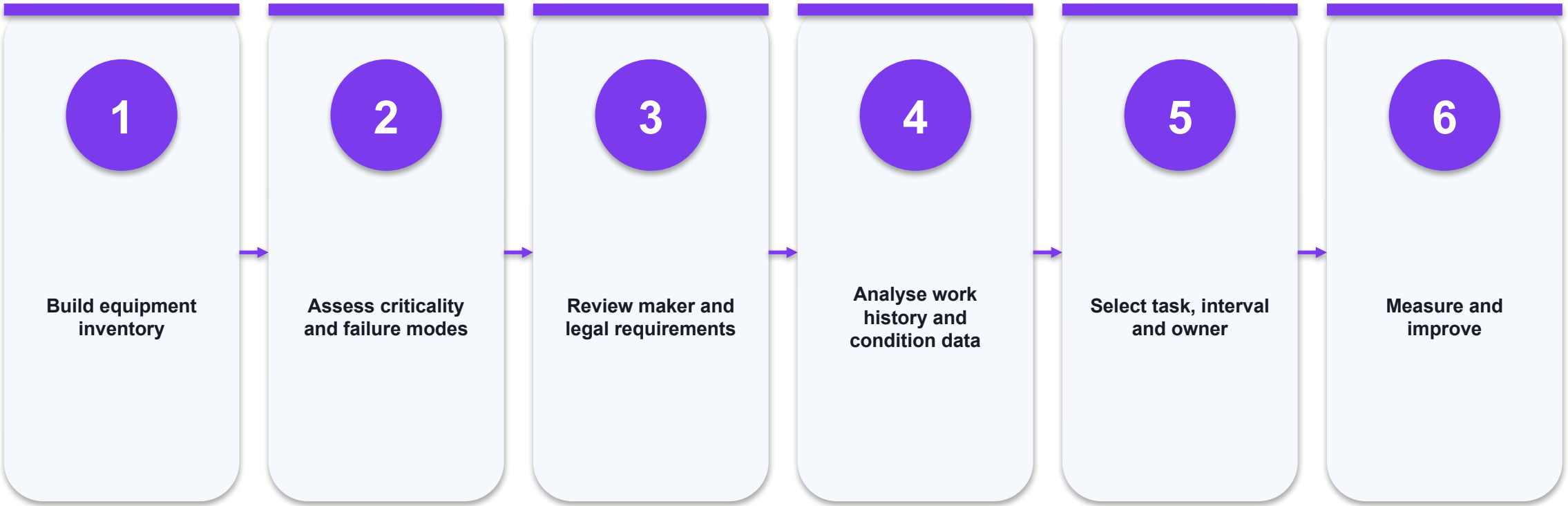
Condition and prediction based

1. Condition-based: threshold or inspection signal
2. Predictive: model estimates risk or remaining life
3. Targets intervention
4. Depends on data quality and model performance

Select the strategy by failure consequence, detectability and economics.

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Determine the maintenance need



Safety-critical maintenance requirements are constraints, not optimisation suggestions.

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FMEA focuses maintenance effort

1 Failure mode

How the asset or process could fail.

2 Effect

Impact on safety, quality, delivery and cost.

3 Cause

Mechanism that creates the failure.

4 Controls

Prevention and detection already in place.

5 Priority

Use severity first, then occurrence and detection evidence.

6 Action

Assign owner, due date and verification evidence.

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Total Productive Maintenance

1 Autonomous maintenance

Operators care for basic conditions and detect abnormalities.

2 Planned maintenance

Schedule evidence-based tasks and spares.

3 Focused improvement

Remove chronic losses.

4 Quality maintenance

Control equipment conditions that create defects.

5 Early equipment management

Design maintainability into new assets.

6 Training

Build operator and maintainer capability.

7 Safety and environment

Eliminate harm and uncontrolled impact.

8 Office TPM

Improve supporting information and administrative flow.

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Overall Equipment Effectiveness

OEE = Availability x Performance x Quality

WORKED EXAMPLE

0.90 x 0.95 x 0.98 = 83.8%

MANAGEMENT INTERPRETATION

Use OEE to expose loss categories, not to rank unlike equipment or pressure operators.

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Metric tree prevents local optimisation

Lagging outcomes

1. Lead time
2. On-time delivery
3. Cost per good unit
4. Customer complaints
5. OEE and first-pass yield

Leading controls

1. Standard-work adherence
2. Changeover completion
3. Maintenance compliance
4. Data-quality score
5. Improvement action closure

Pair each outcome with controllable process measures and an owner.

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Core lean performance measures

1 Flow

Lead time, cycle time, WIP and queue age.

2 Quality

First-pass yield, defect rate, rework and escape rate.

3 Delivery

On-time delivery, schedule adherence and service level.

4 Equipment

Availability, OEE, MTBF and MTTR.

5 People

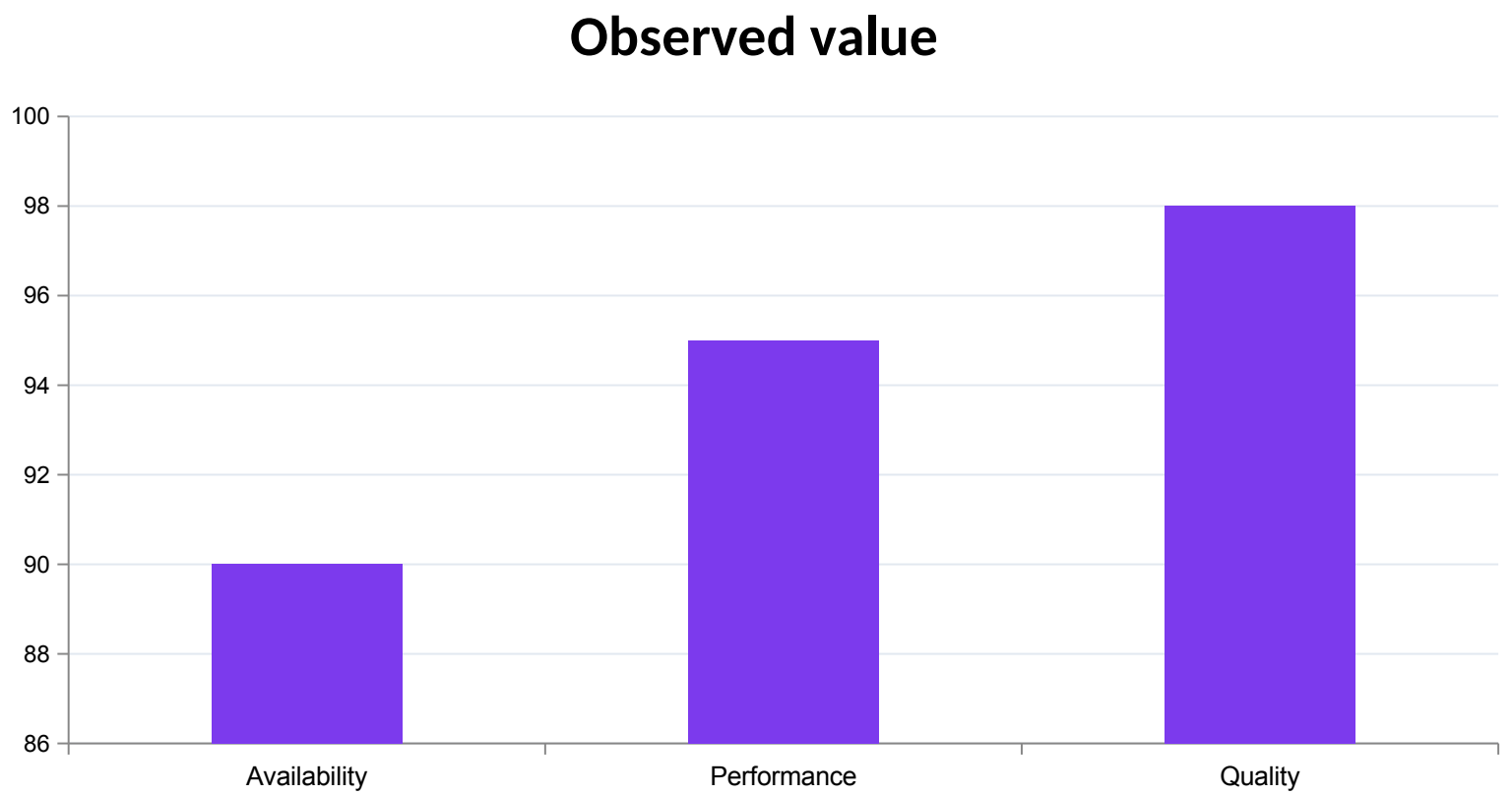
Safety, skills coverage, suggestions and participation.

6 Sustainment

Audit conformance, action closure and benefit retention.

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A KPI dashboard should tell a causal story



WHAT THE DATA SHOWS

High quality does not offset availability loss; target the dominant OEE component and verify the constraint.

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AI use cases in manufacturing

1 Predictive maintenance

Forecast equipment issues from sensor and work-history patterns.

2 Visual quality

Detect, classify or triage defects from images.

3 Process optimisation

Identify bottlenecks and recommend parameter ranges.

4 Demand and inventory

Forecast needs and surface replenishment risk.

5 Digital twins

Simulate assets or flows before physical changes.

6 Copilots

Retrieve manuals, summarise logs and draft controlled work content.

Source: https://www.sciencedirect.com/topics/engineering/lean-manufacturing

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Computer vision quality system

What it can do

1. Consistent high-speed screening
2. Localise visible anomalies
3. Trend defect classes
4. Route uncertain cases

What can go wrong

1. Poor lighting or camera drift
2. Rare defects and class imbalance
3. Product-mix change
4. False negatives released as good
5. Hidden bias in training images

Set disposition authority, confidence thresholds and revalidation triggers.

Source: https://www.sciencedirect.com/topics/engineering/lean-manufacturing

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Predictive-maintenance data path



A model is only as useful as the action workflow it triggers.

Source: https://www.ascm.org/topics/lean-manufacturing/

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Maintenance companion with GenAI

SCENARIO

A technician asks a copilot why a spindle alarm recurs. The system retrieves manuals, approved work instructions and recent work orders.

SIGNALS TO INTERPRET

1. Faster knowledge access
2. Risk of hallucinated torque or sequence
3. Source version may be wrong
4. Maintenance action affects safety

DECISION

Ground responses in approved sources, show citations, require technician verification and never bypass lockout/tagout.

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Generative AI supports knowledge work

1 Retrieve

Find relevant clauses in manuals, standards and work history.

2 Summarise

Condense shift logs, complaints and improvement evidence.

3 Draft

Create first drafts of work instructions, A3s and training notes.

4 Translate

Prepare controlled multilingual instructions for review.

5 Explain

Turn model signals into audience-appropriate decision support.

Generated content remains a draft until an authorised person verifies it.

Source: https://www.ascm.org/topics/lean-manufacturing/

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Classical ML and generative AI

Classical / predictive

1. Outputs scores, classes or forecasts
2. Uses structured signals or labelled examples
3. Evaluate with error metrics and thresholds
4. Good for anomaly, demand, quality and maintenance

Generative

1. Outputs text, images or code
2. Uses prompts plus retrieved context
3. Evaluate groundedness, usefulness and harm
4. Good for knowledge access and drafting

Some solutions combine both: a predictive model signals risk and a grounded copilot explains the evidence.

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Start with manufacturing data

1 Relevance

Use data from the specific asset, process, product and decision.

2 Context

Connect readings with operating state, product mix, maintenance and environment.

3 Quality

Control completeness, accuracy, labels, timing and lineage.

4 Ownership

Assign people accountable for definitions, access and correction.

5 Feedback

Capture decisions and outcomes to learn whether the system helped.

Source: https://mitsloan.mit.edu/ideas-made-to-matter/ai-manufacturing-start-data

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Data readiness gate

Ready enough to pilot

1. Defined outcome and baseline
2. Representative data
3. Known labels and units
4. Access and ownership approved
5. Action workflow exists

Stop and repair

1. No process owner
2. Ambiguous timestamps or units
3. Missing negative cases
4. Uncontrolled personal or sensitive data
5. No way to verify benefit

Do not compensate for poor data with a more complex model.

Source: https://mitsloan.mit.edu/ideas-made-to-matter/ai-manufacturing-start-data

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Six building blocks for manufacturing AI

1 Business application

Prioritised value and use cases.

2 Data

Quality, access, lineage and single source of truth.

3 Technology

Connectivity, platforms, tools and interfaces.

4 Talent and organisation

Domain, data, IT and change capability.

5 Process

Governance, collaboration and delivery method.

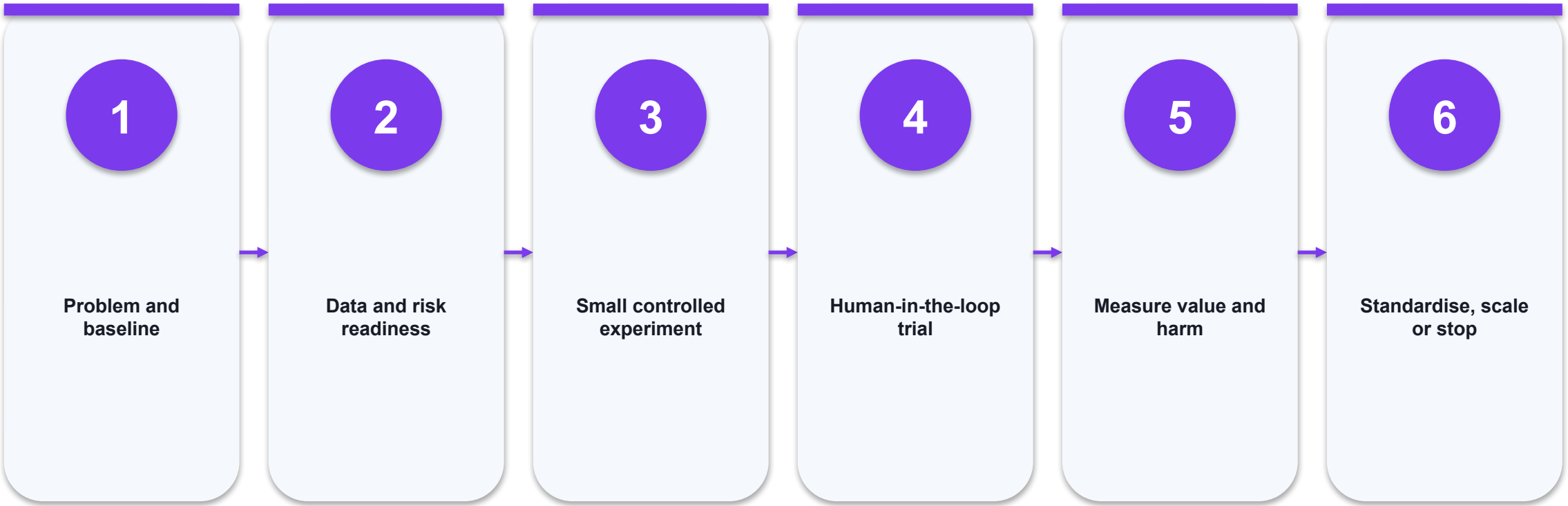
6 Culture

Experimentation, trust and evidence-based decisions.

Source: https://www.pwc.com/gx/en/industrial-manufacturing/pdf/intro-implementing-ai-manufacturing.pdf

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Lean-AI pilot lifecycle



Think big about the operating model, start small with a measurable lighthouse use case.

Source: https://www.pwc.com/gx/en/industrial-manufacturing/pdf/intro-implementing-ai-manufacturing.pdf

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Responsible AI controls on the shop floor

1 Govern

Owner, purpose, approvals, risk class and documentation.

2 Map

Users, affected people, context, failure modes and dependencies.

3 Measure

Accuracy, drift, false alarms, latency, safety and business outcomes.

4 Manage

Thresholds, human review, fallback, monitoring, change control and retirement.

Source: https://www.nist.gov/itl/ai-risk-management-framework

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Human-in-the-loop is a designed control

AI responsibility

1. Detect or predict
2. Present evidence and confidence
3. Recommend within limits
4. Log inputs and outputs

Human responsibility

1. Verify context
2. Approve consequential action
3. Handle exceptions
4. Escalate uncertainty
5. Own safety and customer impact

Name the decision owner and the fallback before deployment.

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Common AI failure modes

1 Data drift

Process, product or environment changes.

2 Automation bias

People accept the recommendation without challenge.

3 Hallucination

Generated content is plausible but unsupported.

4 Feedback loop

Decisions alter future data and reinforce errors.

5 Security

Prompts, data, models or interfaces are manipulated.

6 Hidden transfer

A vendor model changes without local validation.

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When a pilot should stop

SCENARIO

A defect model reaches 96% overall accuracy, but misses two of ten safety-critical crack examples.

SIGNALS TO INTERPRET

1. Overall accuracy hides class-specific harm
2. Safety severity is high
3. Rare class evidence is weak
4. Operator over-trust is plausible

DECISION

Stop automatic disposition; improve data and controls. Use the model only for assistive triage until the critical-class requirement is met.

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AI use-case prioritisation

1 Value

Waste, constraint or risk addressed and size of baseline loss.

2 Feasibility

Data, integration, skills and actionability.

3 Risk

Safety, quality, privacy, security and regulatory consequence.

4 Time to learn

Speed to a controlled, informative experiment.

5 Scalability

Repeatability across assets, products or sites.

6 Adoption

Fit with operator work and decision rights.

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Pilot economics

Benefits

1. Avoided downtime
2. Reduced scrap and rework
3. Lower inspection effort
4. Inventory and lead-time reduction
5. Faster knowledge access

Full cost

1. Data collection and cleaning
2. Integration and validation
3. Change management and training
4. Monitoring, support and model updates
5. Residual risk and fallback capacity

Report net verified benefit and confidence, not vendor headline potential.

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Simple pilot ROI

ROI = (verified annual benefit - annualised total cost) / annualised total cost

WORKED EXAMPLE

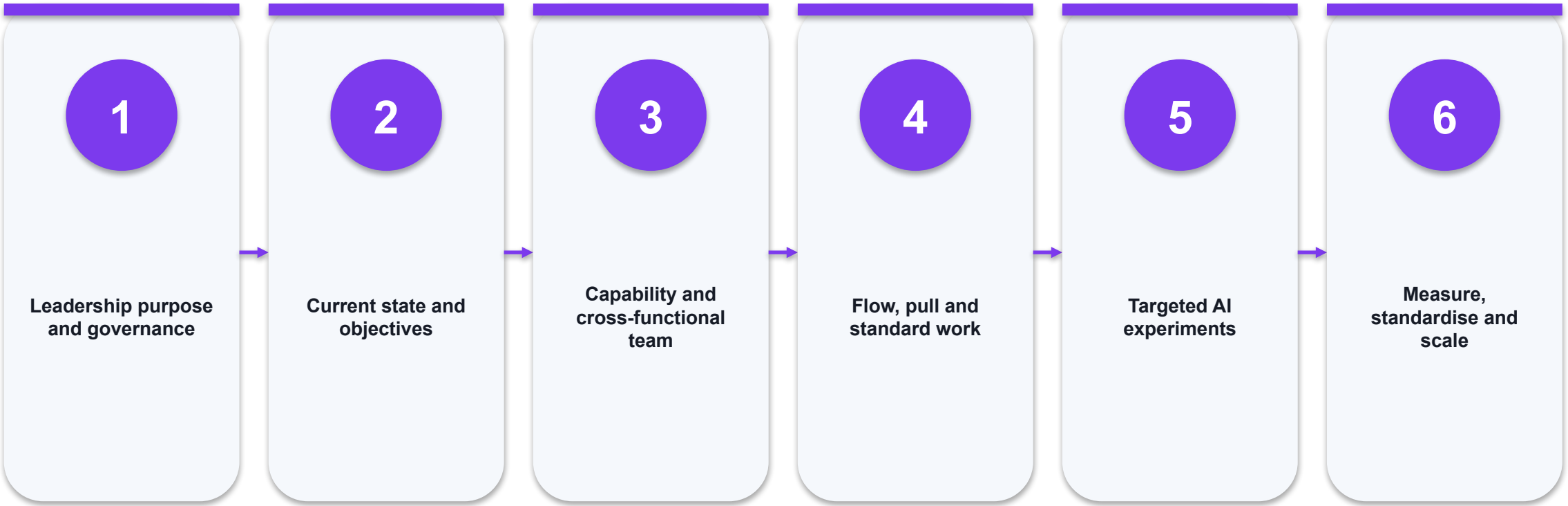
($120,000 - $80,000) / $80,000 = 50%

MANAGEMENT INTERPRETATION

Separate realised from projected benefit and include the cost of controls, not only model licensing.

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Lean implementation sequence



Implementation is a learning system, not a one-time launch.

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Sustaining the new standard

1 Standard work

Document the verified method and decision rights.

2 Visual management

Show current state, targets, exceptions and ownership.

3 Leader routine

Review process evidence at the gemba.

4 Skills

Train operators, engineers, maintainers and managers.

5 Control plan

Define checks, thresholds, reactions and records.

6 Audit and refresh

Confirm benefits, data quality, model performance and change triggers.

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Scaling without copying blindly

SCENARIO

A predictive-maintenance pilot works on one CNC family. Leaders want to deploy it to every machine next month.

SIGNALS TO INTERPRET

1. Sensors and failure modes differ
2. Maintenance history quality varies
3. Action thresholds are asset-specific
4. Support capacity may be insufficient

DECISION

Scale through a repeatable qualification gate: process similarity, data readiness, risk review, local validation and owner acceptance.

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The lean-AI operating principle

1 Lean defines value

Use customer value and system waste as the compass.

2 People own the process

Domain expertise frames, verifies and improves decisions.

3 Data makes reality visible

Quality, context and lineage determine trust.

4 AI shortens the loop

Sense, predict, retrieve or recommend earlier.

5 Governance protects outcomes

Design limits, review, fallback and monitoring.

6 Learning sustains improvement

Standardise verified gains and revisit assumptions.

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ADVANCED | K3 A6

Activity 7: AI Visual Quality Pilot

THE CHALLENGE

Final inspection is a bottleneck. Orion is considering a computer-vision model to flag surface defects, but safety-critical cracks are rare.

PURPOSE

Design a human-in-the-loop visual-quality pilot with suitable data, class metrics, thresholds and fallback.

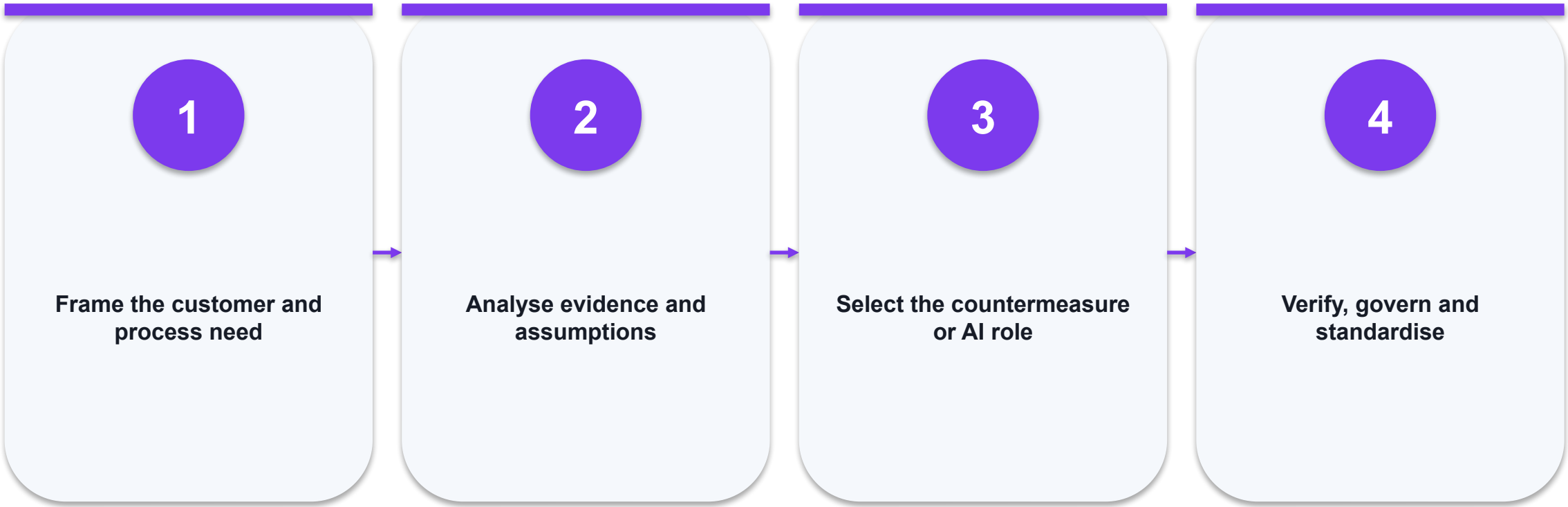
LEARNER OUTPUT

Use-case canvas, data plan, confusion-matrix interpretation, decision rights and pilot acceptance criteria.

Tools: Use-case worksheet; sample predictions CSV; checklist | Detailed procedure: Learner Guide and activity folder

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Activity 7 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 7: evidence and acceptance

1 Question

Why is overall accuracy insufficient?

2 Question

Which error is most harmful?

3 Question

When must a human inspect?

4 Question

What change triggers model revalidation?

5 Acceptance

Metrics are class-specific and reconcile to the CSV.

6 Acceptance

Critical false negatives are treated as a stop condition.

7 Acceptance

Human disposition authority is explicit.

8 Acceptance

Data and environment drift triggers are documented.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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ADVANCED | K3 A5

Activity 8: Predictive-maintenance Decision Design

THE CHALLENGE

CNC-2 shows rising vibration and spindle temperature. A vendor model gives a failure-risk score but does not explain the required maintenance action.

PURPOSE

Combine asset criticality, condition data, work history and model evidence into a controlled maintenance decision.

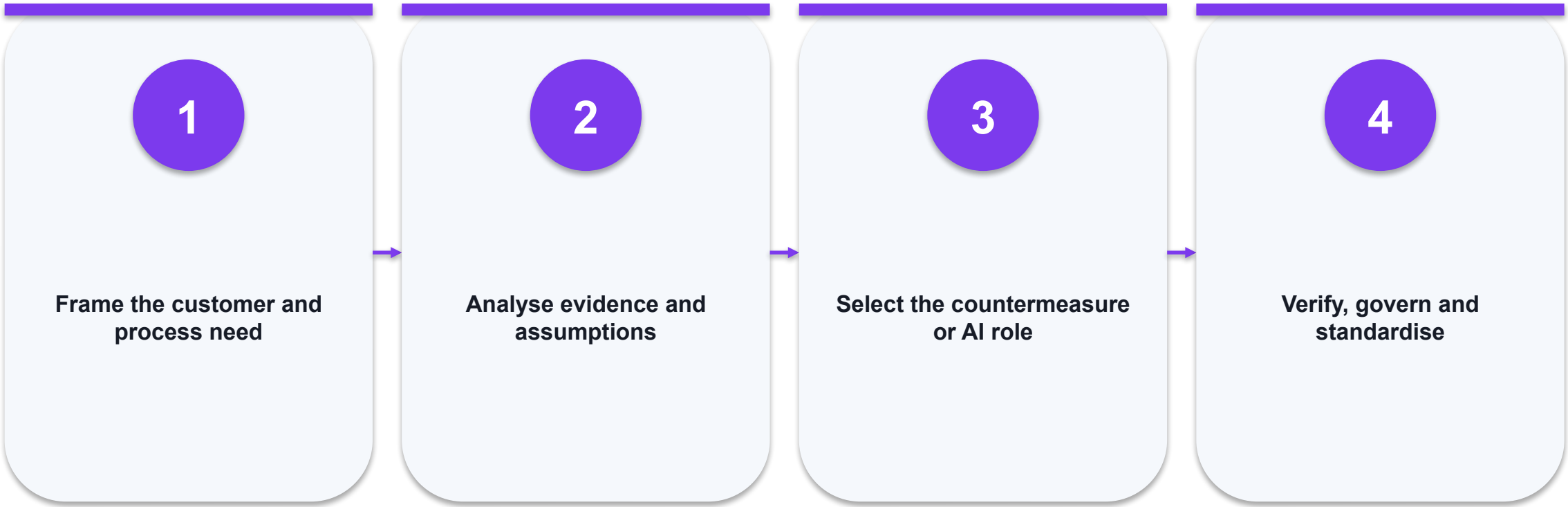
LEARNER OUTPUT

Criticality/FMEA screen, condition trend, action threshold, work-order recommendation and fallback.

Tools: Sensor CSV; FMEA worksheet; maintenance checklist | Detailed procedure: Learner Guide and activity folder

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Activity 8 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 8: evidence and acceptance

1 Question

What failure mode is plausible?

2 Question

Does the trend cross an engineering limit?

3 Question

What action follows each risk band?

4 Question

Who may defer maintenance and on what evidence?

5 Acceptance

Decision uses both engineering and model evidence.

6 Acceptance

Safety and maker requirements cannot be overridden by AI.

7 Acceptance

Every risk band has an owner and action.

8 Acceptance

Fallback maintains a safe maintenance regime.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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ADVANCED | K8 A6

Activity 9: Grounded GenAI for Standard Work

THE CHALLENGE

Maintenance knowledge is split across manuals, approved work instructions and technician notes. Orion wants a copilot to draft troubleshooting guidance.

PURPOSE

Design prompts, retrieval boundaries and review controls that keep generated standard work grounded and safe.

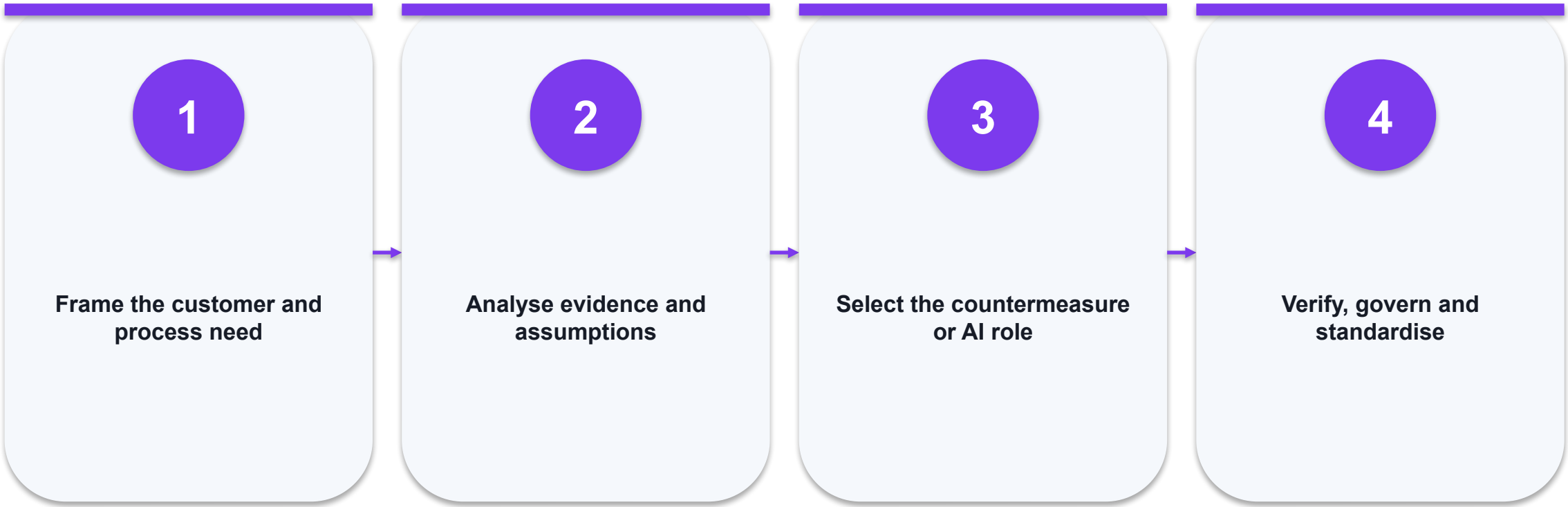
LEARNER OUTPUT

Grounded prompt, source hierarchy, draft instruction, reviewer checklist and controlled-release workflow.

Tools: Prompt worksheet; source register; review checklist | Detailed procedure: Learner Guide and activity folder

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Activity 9 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 9: evidence and acceptance

1 Question

Which sources are authoritative?

2 Question

What must the model refuse to infer?

3 Question

Which fields require exact citation?

4 Question

Who releases the final instruction?

5 Acceptance

Response cites an approved source for every technical instruction.

6 Acceptance

Unknown information is stated as unknown.

7 Acceptance

Safety and lockout/tagout prerequisites are retained.

8 Acceptance

Final release has reviewer, version and effective date.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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CAPSTONE | K1-K8 A1-A6

Activity 10: Lean-AI Pilot Charter and Control Plan

THE CHALLENGE

Orion must choose one improvement: reduce inspection queue, improve CNC uptime or stabilise material replenishment. Each option has different value, data readiness and risk.

PURPOSE

Integrate lean diagnosis, AI use-case selection, implementation planning, governance, economics and sustainment into a defensible pilot charter.

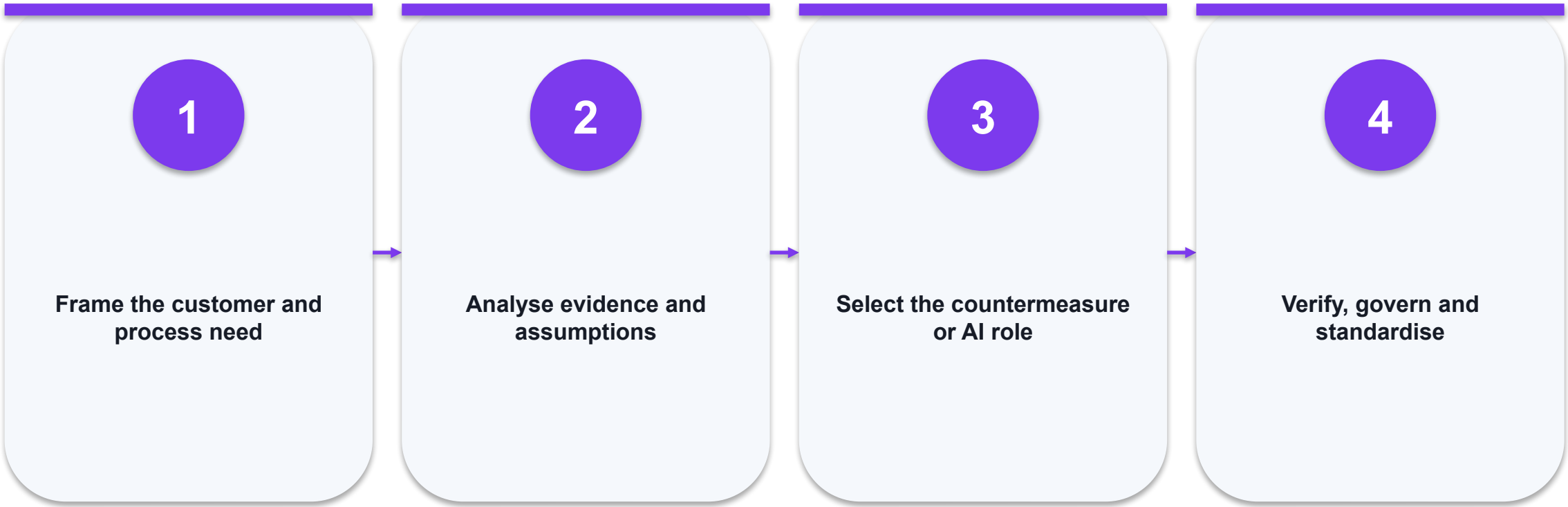
LEARNER OUTPUT

A3/pilot charter, SIPOC, baseline, prioritisation matrix, risk controls, ROI, control plan and executive recommendation.

Tools: SIPOC, system-loop, 5 Whys, Fishbone and Pareto tools; capstone worksheet | Detailed procedure: Learner Guide and activity folder

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Activity 10 decision workflow



Use the detailed numbered procedure in the Learner Guide; the slide stays focused on the decision logic.

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Activity 10: evidence and acceptance

1 Question

Which use case addresses the system constraint?

2 Question

What is the simplest informative experiment?

3 Question

What are the stop conditions?

4 Question

What evidence supports scale, redesign or termination?

5 Acceptance

Selected use case addresses an evidenced system need.

6 Acceptance

Charter includes baseline, target, scope, owner and timebox.

7 Acceptance

Human authority, fallback and stop criteria are explicit.

8 Acceptance

ROI distinguishes projected and verified benefit.

Submit the completed worksheet, calculations and evidence checklist from the activity folder.

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Topic 3 recap

1 Knowledge

Steps required for successful lean implementation

2 Ability

K3 K8 A5 A6

3 Evidence

Use the completed activity outputs to explain your decision.

4 Transfer

Name one workplace process where the method applies.

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Tea Break

10 minutes | Finalise your capstone evidence

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COURSE SUMMARY

What you achieved

1 See waste

Classified eight wastes and framed problems from evidence.

2 Design flow

Used value, flow, pull, takt, ABC, EBQ and kanban.

3 Protect reliability

Designed TPM and maintenance decisions.

4 Apply AI

Matched predictive, vision and GenAI capabilities to lean needs.

5 Govern decisions

Defined data, human authority, fallback and monitoring.

6 Scale learning

Built a measurable pilot charter and control plan.

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CONTINUE LEARNING

Support and course resources

1 LMS

https://lms-tms.tertiaryinfotech.com/

2 Course page

https://www.tertiarycourses.com.sg/wsq-reducing-waste-and-improving-workplace-efficiency-with-lean-six-sigma.html

3 Email

enquiry@tertiaryinfotech.com

4 Telephone

+65 6100 0613

Use approved workplace data only and preserve your organisation's confidentiality.

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FINAL ASSESSMENT

Assessment reminder

1 WA

8 open-ended questions | 60 minutes | K1-K8.

2 PP

6 practical tasks | 90 minutes | A1-A6.

3 Open book

Slides, Learner Guide and approved materials only.

4 Submit

Upload completed documents to the LMS and retain confirmation.

Take the assessment digital attendance before starting.

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Assessment flow



If you disagree with the outcome, do not sign acceptance; follow the appeal process.

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Digital Attendance (Mandatory)

1 Scan

Use the assessment digital-attendance QR code shown by the trainer.

2 Verify

Confirm the submission on your device.

3 Complete

Keep attendance separate from assessment submission.

4 Ask

Tell the trainer immediately if the attendance event fails.

This slide is repeated at the end as required.

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THANK YOU

Make the process visible. Improve the system. Use AI responsibly.

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